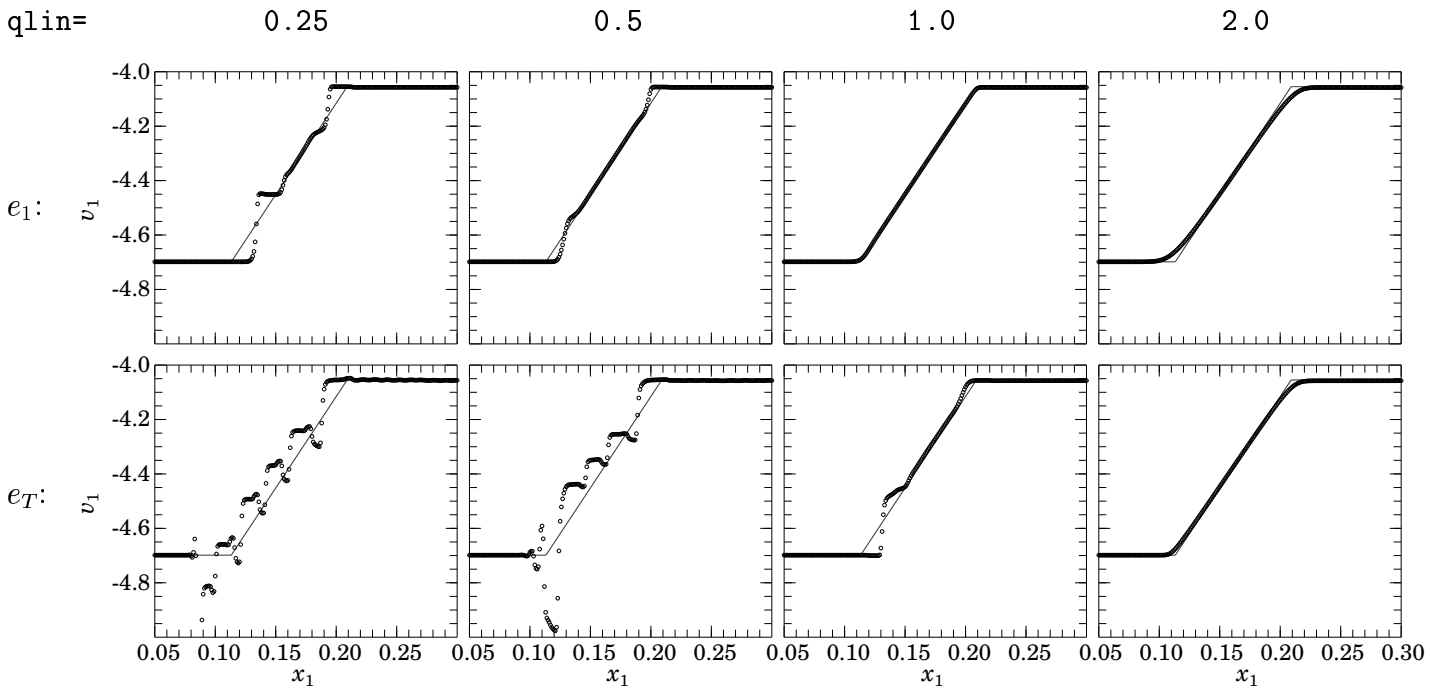
ZEUS-3D 1-D Gallery #14: Rarefaction Shocks



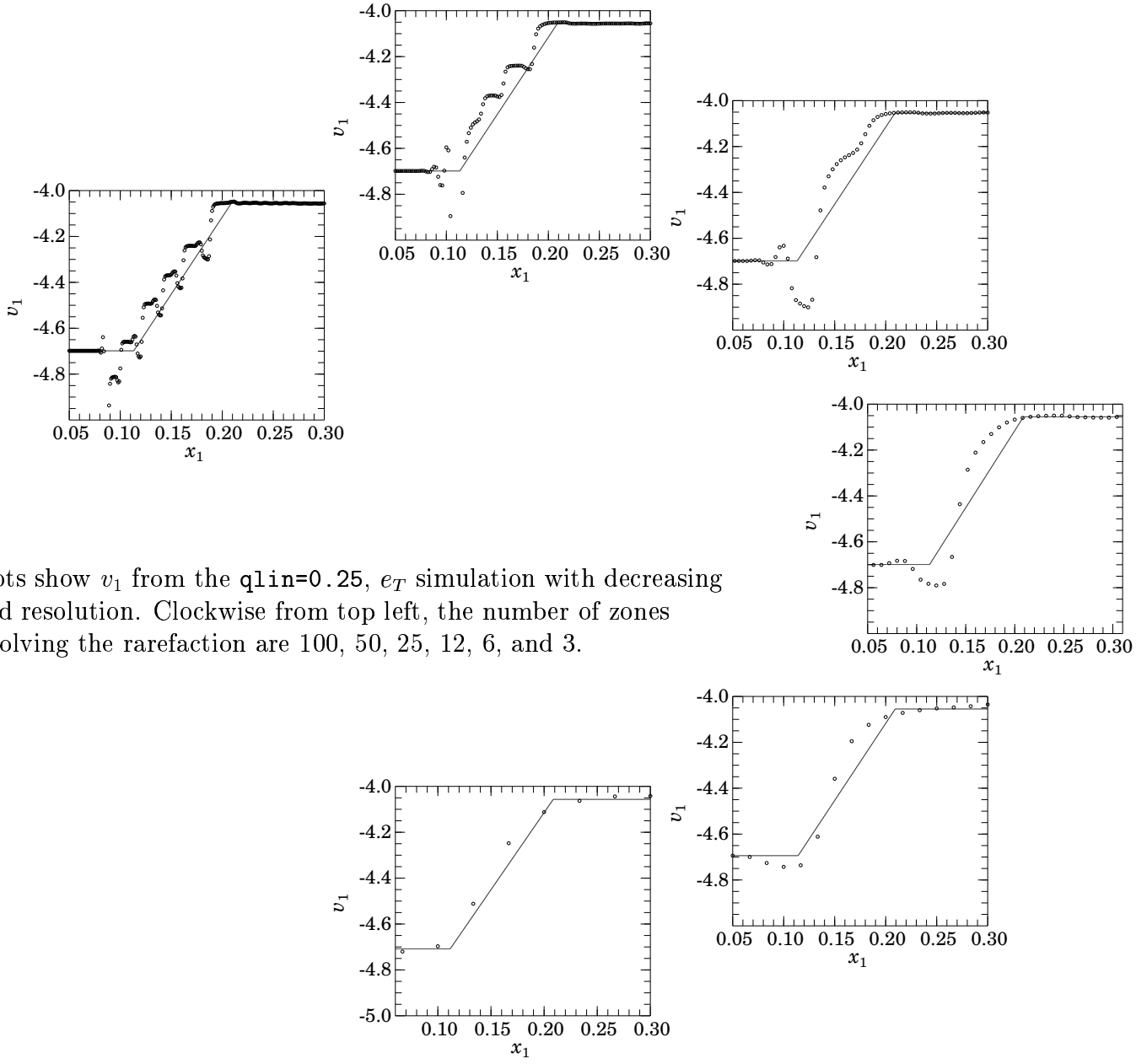
This study is based on Fig. 2 from Falle (2002, ApJ, 577, L123), showing the solution of the MHD shock tube test with the left state $(\rho, v_1, v_2, v_3, B_2, B_3, p_1) = (1, -4.6985, -1.085146, 0, 1.9680, 0, 0.2327)$ and the right state $(0.7270, -4.0577, -0.8349, 0, 1.355, 0, 0.1368)$ with $B_1 = -0.7$, $\gamma = 5/3$ at time $t = 0.1$. At $t = 0$, the discontinuity is at $x_1 = 0.8$. The grid domain is $0 \le x_1 \le 1$, though only the portion of the grid containing the rarefaction is shown.

Open circles are the `dzeus35` solution for v_1 using 1,000 zones, `CMoC`, the internal/total energy equation on the top/bottom row (labelled e_1/e_T), and second-order interpolation. `dzeus35` parameters controlling the time step and artificial viscosity are: `courno=0.5`, `qcon=0.0`, and `qlin` as indicated at the top of each column. Lines are the results from the non-linear Riemann solver described in Ryu & Jones (1995, ApJ, 442, 228).

The same issues as discussed for the hydrodynamical rarefaction in problem #13 in the 1-D Gallery apply to this MHD rarefaction too, though the linear viscosity required to stablise this rarefaction and prevent rarefaction shocks is considerably higher. According to Falle (2002), this problem exemplifies the imbalanced order of the scheme (second order space, first order time because of the operator split momentum equation). As noted in the document *What is ZEUS-3D?*, the momentum equation will not be operator split in the next release of the code and, should Falle's conclusion be correct, this should help cure the rarefaction shock problem.

As noted by Falle (2002), the appearance of rarefaction shocks depends as much on resolution and frame of reference as on linear viscosity. The sequence of images on the next page show, for a fixed `qlin` (`0.25`), `dzeus35` does better at representing the rarefaction the fewer zones it has to resolve it. In a full 3-D simulation, one rarely has the luxury of resolving individual MHD waves with 100 zones (as in the lead image of the sequence), and the normal requirement is that such features be resolved with a few or several zones (as in the last image of the sequence).

Finally, at the bottom of the next page is the full-resolution problem done under a Galilean transformation ($v_1' = v_1 + 4$) that renders both the pre- and post-rarefaction speeds sub-Alfvénic. In this case, `qlin=0.5` allows the rarefaction to be represented reasonably well.



Plots show v_1 from the `qlin=0.25`, e_T simulation with decreasing grid resolution. Clockwise from top left, the number of zones resolving the rarefaction are 100, 50, 25, 12, 6, and 3.

Same as the `qlin=0.5`, e_T simulation, but with $v_{1L} = -0.6985$ and $v_{1R} = -0.0577$ (thus, a Galilean transformation of $\Delta v_1 = 4$ has been applied). In this frame of reference, the rarefaction fan is well-represented with a much smaller linear viscosity.

